for the age of the visitors (based on their registration details) and derived the quartiles from that. Maybe we used Excel, Python, or `d3.scale.quartile()`, or maybe it was part of a dataset we downloaded. As you work with data, you'll be exposed to common statistical summaries like this and you'll have to represent them as part of your charts, so don't be too intimidated by it. We'll use a CSV format for the information.

The following listing shows our dataset with the number of registered users that visit the site each day, and the quartiles of their ages.

Listing 4.2 boxplots.csv

```
day,min,max,median,q1,q3,number
1,14,65,33,20,35,22
2,25,73,25,25,30,170
3,15,40,25,17,28,185
4,18,55,33,28,42,135
5,14,66,35,22,45,150
6,22,70,34,28,42,170
7,14,65,33,30,50,28
```

When we map the median age as a scatterplot, as in figure 4.12, it looks like there's not too much variation in our user base throughout the week. We do that by drawing scatterplot points for each day at the median age of the visitor for that day. We'll also invert the y-axis so that it makes a bit more sense.

Listing 4.3 Scatterplot of average age

```
    d3.csv("boxplot.csv", scatterplot)

function scatterplot(data) {
    xScale = d3.scale.linear().domain([1,8]).range([20,470]);
    yScale = d3.scale.linear().domain([0,100]).range([480,20]);

    yAxis = d3.svg.axis()
         .scale(yScale)
         .orient("right")
         .ticks(8)
         .tickSize(-470);

    d3.select("svg").append("g")
         .attr("transform", "translate(470,0)")
         .attr("id", "yAxisG")
         .call(yAxis);

    xAxis = d3.svg.axis()
         .scale(xScale)
         .orient("bottom")
         .tickSize(-470)
         .tickValues([1,2,3,4,5,6,7]);

    d3.select("svg").append("g")
         .attr("transform", "translate(0,480)")
         .attr("id", "xAxisG")
         .call(xAxis);
```

Scale is inverted, so higher values are drawn higher up and lower values toward the bottom

Offsets the <g> containing the axis

Specifies the exact tick values to correspond with the numbered days of the week